Descriptive Set Theory

MATH 592 · Scribed Course Notes · Part 2: Definable Subsets of Polish Spaces

Definitions Results
Examples Notes
green text marks a term or notation at the point where it is defined

Contents

1 Course Overview **1**

2 σ-Algebras and Borel Sets **1**
2.1 σ-Algebras, Measurable Spaces and Borel Sets 2
2.2 Stratification of Borel Sets into a Hierarchy 3
2.3 The Classes $\mathcal{B}$, $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$ 7
2.4 Universal Sets and Diagonalization 9

3 Change of Topology **13**
3.1 Turning Borel Sets into Clopen Sets 13
3.2 Making Closed-Graph Functions Continuous for Free 16

4 Analytic Sets **18**
4.1 Definition and Closure Properties 18
4.2 Universal Sets for Analytic / Coanalytic Sets 19
4.3 Analytic Separation 20
4.4 Souslin's Operation $\mathcal{A}$ 22

Source manuscript page 1

1 Course Overview

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Note 1. The abbreviations used in the manuscript have been expanded in this edition.

Note 2. Text set in blue (or red) reproduces the scribe's own annotations, written in blue (or red) ink in the manuscript.

Note 3. The marker "Source manuscript page n" indicates where page n of the handwritten manuscript (Part 2) begins.

Note 4. Following the colour convention of the lecturer's notes, a term or a piece of notation is set in green at the point where it is defined, e.g. "a set is analytic if ..." or "$\mathcal{B}(X) := \dots$". Italics are reserved for ordinary emphasis.

Note 5. Results are numbered within sections, and a reference such as Theorem 3.1 is a clickable link to the result it cites; the number is set in this colour so that links are recognizable. Section references like §2.4 are links too.

Note 6. Results from Part 1 are cited by their number in the Part 1 edition of these notes, e.g. Part 1, Theorem 2.9 (Alexandrov's theorem). These are links into the file `Math592_Part1_Polish_Spaces.pdf`, which work when the two PDF files are kept in the same folder. Homework problems ("HW 3, Q2") are not part of the notes and are cited by name only.

Note 7. The manuscript ends with the first two pages of a section on uniformization theorems (manuscript pp. 42–43), which is still in progress; it is not included in this edition.

Relevant readings as prerequisites.

- Analysis 1: § Topology and definitions.
- Analysis 2: § Metric spaces.
- Folland, *Real Analysis*, Chapter 4.

Source manuscript page 2

Introduction. In Part 1, we discussed the properties of Polish spaces and their characterizations. In this part, we are mainly concerned with the properties of subsets of Polish spaces, and of functions between Polish spaces, putting our focus on the so-called "definability".

We begin our discussion with Borel measurability and Borel (measurable) sets and functions.

Source manuscript page 3

2 σ-Algebras and Borel Sets

2.1 σ-Algebras, Measurable Spaces and Borel Sets

Definition 2.1

Recall that an algebra on a given set X is a collection of subsets $\mathcal{A} \subseteq \mathcal{P}(X)$ which contains $\varnothing$ and is closed under complements and finite unions (hence also finite intersections). A σ-algebra is an algebra which is closed under countable unions (hence also countable intersections). □

Remark 2.2

Arbitrary intersections of σ-algebras on X are still σ-algebras, so for a given collection $\mathcal{E} \subseteq \mathcal{P}(X)$, it makes sense to define the σ-algebra generated by $\mathcal{E}$ as the intersection of all σ-algebras containing $\mathcal{E}$. In other words, this is the $\subseteq$-smallest σ-algebra containing $\mathcal{E}$, denoted $\langle \mathcal{E} \rangle_\sigma$, and $\langle \mathcal{E} \rangle$ denotes the algebra generated by $\mathcal{E}$.

For $\mathcal{E} \subseteq \mathcal{P}(X)$, denote $\neg\mathcal{E} := \{X \setminus A : A \in \mathcal{E}\}$, the dual collection of $\mathcal{E}$. □

Proposition 2.3

Let X be a set and $\varnothing \in \mathcal{E} \subseteq \mathcal{P}(X)$. Then $\langle \mathcal{E} \rangle_\sigma$ is the $\subseteq$-smallest collection $\mathcal{S}$ which contains $\mathcal{E} \sqcup \neg\mathcal{E}$ and is closed under countable unions and countable intersections.

Proof. Put $\widetilde{\mathcal{S}} := \{A \in \mathcal{S} : A, A^c \in \mathcal{S}\} \subseteq \mathcal{S}$. We show $\widetilde{\mathcal{S}}$ is a σ-algebra containing $\mathcal{E}$, hence $\widetilde{\mathcal{S}} \supseteq \langle \mathcal{E} \rangle_\sigma$, and $\widetilde{\mathcal{S}} \subseteq \mathcal{S} \subseteq \langle \mathcal{E} \rangle_\sigma$ (the latter since $\langle \mathcal{E} \rangle_\sigma$ is itself a collection with the two properties, and $\mathcal{S}$ is the smallest one), hence $\widetilde{\mathcal{S}} = \mathcal{S} = \langle \mathcal{E} \rangle_\sigma$.

Observe that $\mathcal{E} \subseteq \widetilde{\mathcal{S}}$ since $A \in \mathcal{E} \implies A^c \in \neg\mathcal{E}$, and $\mathcal{S} \supseteq \mathcal{E} \sqcup \neg\mathcal{E} \implies A, A^c \in \mathcal{S} \implies A \in \widetilde{\mathcal{S}}$. Moreover, $\widetilde{\mathcal{S}}$ is a σ-algebra: For $A \in \widetilde{\mathcal{S}}$, $A^c \in \widetilde{\mathcal{S}}$ by definition, and

Source manuscript page 4

for $(A_n)_{n\in\mathbb{N}} \subseteq \widetilde{\mathcal{S}}$, $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{S}$ since $\mathcal{S}$ is closed under countable unions, and $\left(\bigcup_{n\in\mathbb{N}} A_n\right)^c = \bigcap_{n\in\mathbb{N}} A_n^c \in \mathcal{S}$ since $\mathcal{S}$ is closed under countable intersections; thus $\bigcup_{n\in\mathbb{N}} A_n \in \widetilde{\mathcal{S}}$. Lastly $\varnothing \in \widetilde{\mathcal{S}}$. Hence $\mathcal{E} \subseteq \widetilde{\mathcal{S}} \implies \langle \mathcal{E} \rangle_\sigma \subseteq \widetilde{\mathcal{S}}$. □

Definition 2.4

A measurable space is a set X equipped with a σ-algebra $\mathcal{I}$, i.e. a tuple $(X, \mathcal{I})$. For measurable spaces $(X, \mathcal{I})$, $(Y, \mathcal{J})$, a function $f : X \to Y$ is called measurable or $(\mathcal{I}, \mathcal{J})$-measurable if f^{-1} takes sets in $\mathcal{J}$ to sets in $\mathcal{I}$, i.e. $f^{-1}(B) \in \mathcal{I}$, $\forall B \in \mathcal{J}$.

When X is a topological space, the σ-algebra generated by the collection of all open sets is called the Borel σ-algebra of X, denoted $\mathcal{B}(X)$, and the sets in $\mathcal{B}(X)$ are called Borel sets.

For a function $f : X \to Y$ from a measurable space $(X, \mathcal{I})$ to a topological space Y, we say that f is measurable or $\mathcal{I}$-measurable if f is $(\mathcal{I}, \mathcal{B}(Y))$-measurable. Lastly, if X, Y are topological spaces, then a function $f : X \to Y$ is called Borel if it is $(\mathcal{B}(X), \mathcal{B}(Y))$-measurable. □

Note 2.5

If we ask for $(\mathcal{I}, \mathcal{J})$-measurability instead of $(\mathcal{I}, \mathcal{B}(Y))$-measurability (for a σ-algebra $\mathcal{J}$ on Y larger than $\mathcal{B}(Y)$), then the class of functions will be much smaller and less useful. □

Proposition 2.6

Continuous functions are Borel. More generally, let

Source manuscript page 5

$(X, \mathcal{I})$, $(Y, \mathcal{J})$ be measurable spaces, $\mathcal{E} \subseteq \mathcal{J}$ generating $\mathcal{J}$, i.e. $\langle \mathcal{E} \rangle_\sigma = \mathcal{J}$. Then if $f : X \to Y$ satisfies $f^{-1}(B) \in \mathcal{I}$, $\forall B \in \mathcal{E}$, then f is $(\mathcal{I}, \mathcal{J})$-measurable.

Proof. Note that the second part implies the first part. Define $\mathcal{J}_0 := \{B \in \mathcal{J} : f^{-1}(B) \in \mathcal{I}\}$. Then $\mathcal{J}_0 \subseteq \mathcal{J}$ while $\mathcal{J}_0 \supseteq \mathcal{E}$. Furthermore, $\mathcal{J}_0$ is a σ-algebra since $\varnothing = f^{-1}(\varnothing) \in \mathcal{I}$, and for $B, B_0, B_1, \ldots \in \mathcal{J}_0$, $f^{-1}(B^c) = f^{-1}(B)^c \in \mathcal{I}$, $f^{-1}(\bigcup_n B_n) = \bigcup_n f^{-1}(B_n) \in \mathcal{I}$, i.e. f^{-1} commutes with these operations and $\mathcal{I}$ is closed under them. Hence $\mathcal{J}_0 = \mathcal{J} = \langle \mathcal{E} \rangle_\sigma$. □

2.2 Stratification of Borel Sets into a Hierarchy

Definition 2.7

Let X be a topological space. We define the hierarchy of Borel sets level-by-level, via transfinite recursion. Let ω_1 denote the first uncountable ordinal. For each $1 \leq \xi < \omega_1$, define the collections of subsets of X as follows:

$$\mathbf{\Sigma}^0_1(X) := \{U \subseteq X : U \text{ open}\}, \qquad \mathbf{\Pi}^0_\xi(X) := \neg\mathbf{\Sigma}^0_\xi(X),$$
$$\mathbf{\Sigma}^0_\xi(X) := \Big\{ \bigcup_{n\in\mathbb{N}} A_n : A_n \in \mathbf{\Pi}^0_{\xi_n}(X),\ \xi_n < \xi,\ \forall n \in \mathbb{N} \Big\} \quad (2 \leq \xi < \omega_1).$$

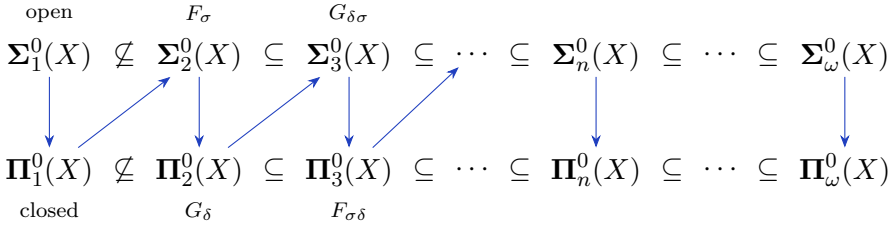
In addition, we define the so-called ambiguous classes:

$$\mathbf{\Delta}^0_\xi(X) := \mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X). \quad \square$$

Observation. Notice that by definition, for $2 \leq \xi < \eta$, $\mathbf{\Sigma}^0_\xi(X) \subseteq \mathbf{\Sigma}^0_\eta(X)$, since the collection we can pick the A_n's from is enlarged from ξ to η, and $\mathbf{\Pi}^0_\xi(X) = \neg\mathbf{\Sigma}^0_\xi(X) \subseteq \neg\mathbf{\Sigma}^0_\eta(X) = \mathbf{\Pi}^0_\eta(X)$.

Source manuscript page 6

Hence we have the following diagram:



(Blue arrows: each $\mathbf{\Pi}^0_\xi$ is the dual of $\mathbf{\Sigma}^0_\xi$, and $\mathbf{\Sigma}^0_{\xi+1}$ consists of the countable unions of $\mathbf{\Pi}^0_\xi$ sets.) Here ω denotes the first infinite ordinal, i.e. $\mathbb{N}$ (ordered as usual). $\mathbf{\Sigma}^0_\omega(X)$ contains unions $\bigcup_{n\in\mathbb{N}} A_n$ where $A_n \in \mathbf{\Pi}^0_n(X)$, and there is no reason why $\bigcup_{n\in\mathbb{N}} A_n$ should appear in $\mathbf{\Sigma}^0_m(X)$, for some $m \in \mathbb{N}$. Thus to obtain the full Borel sets, we must continue transfinitely via the definition above (up to ω_1, as we will see).

In particular, $\mathbf{\Sigma}^0_1(X) \subseteq \mathbf{\Sigma}^0_2(X)$ is not true in general topological spaces, as open sets are not necessarily F_σ; symmetrically, $\mathbf{\Pi}^0_1(X) \subseteq \mathbf{\Pi}^0_2(X)$ is not true in general, as closed sets are not necessarily G_δ. □

Proposition 2.8

Let X be a metrizable space.

(a) $\mathbf{\Sigma}^0_\eta(X) \cup \mathbf{\Pi}^0_\eta(X) \subseteq \mathbf{\Delta}^0_\xi(X)$, $\forall\, 1 \le \eta < \xi < \omega_1$.
(b) $\mathcal{B}(X) = \bigcup_{1\le\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) = \bigcup_{1\le\xi<\omega_1} \mathbf{\Pi}^0_\xi(X) = \bigcup_{1\le\xi<\omega_1} \mathbf{\Delta}^0_\xi(X)$.

Proof. (a) Take $A \in \mathbf{\Sigma}^0_\eta(X)$. Then $A \in \mathbf{\Sigma}^0_\xi(X)$ by our observation above. In particular, $\mathbf{\Sigma}^0_1(X) \subseteq \mathbf{\Sigma}^0_2(X)$ since open sets are F_σ in metrizable spaces (lecture 1, Part 1, Lemma 2.8). $A \in \mathbf{\Pi}^0_\xi(X)$ by taking $A_0 = A \in \mathbf{\Sigma}^0_\eta(X)$, $A_n = X \in \mathbf{\Sigma}^0_1(X)$ for $n \ge 1$; then by definition, $A = \bigcap_{n\in\mathbb{N}} A_n \in \mathbf{\Pi}^0_\xi(X)$.

Source manuscript page 7

Hence $A \in \mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X) = \mathbf{\Delta}^0_\xi(X)$. Lastly, the dual simply takes the complement of the sets in the collection, thus

$$\begin{aligned}\mathbf{\Pi}^0_\eta(X) = \neg\mathbf{\Sigma}^0_\eta(X) &\subseteq \neg(\mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X)) \\ &= (\neg\mathbf{\Sigma}^0_\xi(X)) \cap (\neg\mathbf{\Pi}^0_\xi(X)) = \mathbf{\Pi}^0_\xi(X) \cap \mathbf{\Sigma}^0_\xi(X) \\ &= \mathbf{\Delta}^0_\xi(X).\end{aligned}$$

Hence $\mathbf{\Sigma}^0_\eta(X) \cup \mathbf{\Pi}^0_\eta(X) \subseteq \mathbf{\Delta}^0_\xi(X)$.

(b) Notice that (a) implies that we have an intertwined chain of classes, and consequently

$$\bigcup_{\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) = \bigcup_{\xi<\omega_1} \mathbf{\Pi}^0_\xi(X) = \bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X) :$$

By definition of $\mathbf{\Delta}^0_\xi(X)$, $\bigcup_{\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) \supseteq \bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X)$ and $\bigcup_{\xi<\omega_1} \mathbf{\Pi}^0_\xi(X) \supseteq \bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X)$. On the other hand, by (a),

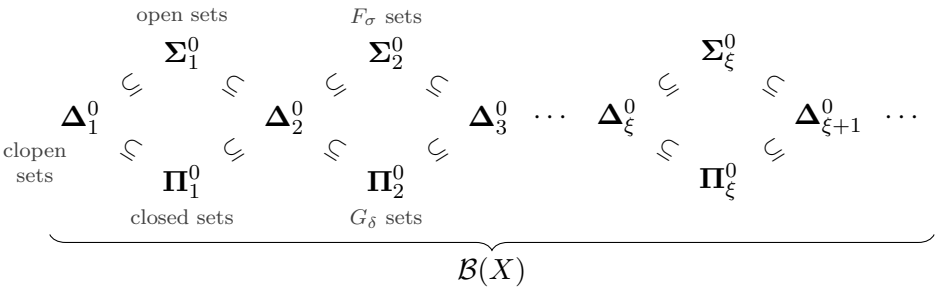
$$\begin{aligned}\bigcup_{\xi<\omega_1} \mathbf{\Delta}^0_\xi(X) \supseteq \bigcup_{1<\xi<\omega_1} \mathbf{\Delta}^0_\xi(X) &\supseteq \bigcup_{\eta<\omega_1} (\mathbf{\Sigma}^0_\eta(X) \cup \mathbf{\Pi}^0_\eta(X)) \\ &= \bigcup_{\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X) \;\cup\; \bigcup_{\xi<\omega_1} \mathbf{\Pi}^0_\xi(X).\end{aligned}$$

It is left to show that $\mathcal{B}(X) =$ any of the unions. Clearly $\mathbf{\Sigma}^0_\xi(X) \subseteq \mathcal{B}(X)$ since at every inductive step we only take complements and countable unions. And $\mathbf{\Sigma}^0_1(X)$ generates $\mathcal{B}(X)$. Hence it remains to show the unions are σ-algebras. Indeed, closure under complement follows from $\neg\mathbf{\Delta}^0_\xi(X) = \mathbf{\Delta}^0_\xi(X)$, and closure under countable union follows from each $\mathbf{\Sigma}^0_\xi(X)$ being closed under countable union (each set in $\mathbf{\Sigma}^0_\xi(X)$ is a countable union of $A_n \in \mathbf{\Pi}^0_{\xi_n}(X)$,

Source manuscript page 8

and their countable union is a countable union of countable unions, hence still a countable union), together with the fact that countably many sets $A_n \in \mathbf{\Sigma}^0_{\xi_n}(X)$ all lie in one $\mathbf{\Sigma}^0_\xi(X)$, namely for $\xi := \sup_n \xi_n < \omega_1$ (a countable supremum of countable ordinals is countable). □

Observation. Thus for metrizable spaces, we have the following picture:



From HW 1, every second countable space has $\leq |2^{\mathbb{N}}| = \mathfrak{c}$ many open sets, hence also closed sets. Taking countable unions of sets from a collection of cardinality $\leq \mathfrak{c}$ still results in $\leq |2^{\mathbb{N}}|$ many sets since $(2^{\mathbb{N}})^{\mathbb{N}} \cong 2^{\mathbb{N}\times\mathbb{N}} \cong 2^{\mathbb{N}}$. Thus by transfinite induction on ξ, $|\mathbf{\Sigma}^0_\xi(X)| \leq \mathfrak{c}$ and $|\mathbf{\Pi}^0_\xi(X)| \leq \mathfrak{c}$. Finally

$$|\mathcal{B}(X)| = \Big|\bigcup_{1\leq\xi<\omega_1} \mathbf{\Sigma}^0_\xi(X)\Big| \leq |\omega_1| \times |2^{\mathbb{N}}| \leq |2^{\mathbb{N}} \times 2^{\mathbb{N}}| = |2^{\mathbb{N}}|,$$

with the last equality given by the interleaving map $((a_n)_n, (b_n)_n) \mapsto (a_1, b_1, a_2, b_2, \dots)$. Hence there are at most continuum-many Borel sets in a second countable space, and continuum is preserved along the zigzag chain above. In contrast, $\mathcal{L}$:= the Lebesgue measurable subsets of $\mathbb{R}$ has cardinality at least $2^{\mathfrak{c}} = 2^{|2^{\mathbb{N}}|}$,

Source manuscript page 9

since the $\frac{1}{3}$-Cantor set has cardinality continuum with Lebesgue measure 0, thus all of its subsets are Lebesgue measurable with measure 0. □

Proposition 2.9

Let X be a topological space and $1 \leq \xi < \omega_1$, i.e. ξ a countable ordinal.

(a) $\mathbf{\Sigma}^0_\xi(X)$ is closed under taking countable unions; $\mathbf{\Pi}^0_\xi(X)$ is closed under taking countable intersections; $\mathbf{\Delta}^0_\xi(X)$ is closed under taking complements.

(b) If X is metrizable, then all three classes are closed under taking finite unions and intersections.

Proof. (a) $\mathbf{\Sigma}^0_\xi(X)$ is closed under taking countable unions since for $\xi = 1$, a countable union of open sets is still open; for $\xi \geq 2$, a countable union of countable unions is still a countable union. $\mathbf{\Pi}^0_\xi(X)$ follows by taking complements with $\mathbf{\Sigma}^0_\xi(X)$, and

$$\neg\mathbf{\Delta}^0_\xi(X) = \neg(\mathbf{\Sigma}^0_\xi(X) \cap \mathbf{\Pi}^0_\xi(X)) = \neg\mathbf{\Sigma}^0_\xi(X) \cap \neg\mathbf{\Pi}^0_\xi(X) = \mathbf{\Pi}^0_\xi(X) \cap \mathbf{\Sigma}^0_\xi(X) = \mathbf{\Delta}^0_\xi(X).$$

(b) It suffices to show $\mathbf{\Sigma}^0_\xi(X)$ is closed under finite intersections and then everything else follows. Take $\bigcup_{n\in\mathbb{N}} A_n, \bigcup_{n\in\mathbb{N}} B_n \in \mathbf{\Sigma}^0_\xi(X)$. Then $\big(\bigcup_{n\in\mathbb{N}} A_n\big) \cap \big(\bigcup_{n\in\mathbb{N}} B_n\big) = \bigcup_{n,m\in\mathbb{N}}(A_n \cap B_m)$, where $A_n \in \mathbf{\Pi}^0_{\xi_n}(X)$,

$B_m \in \mathbf{\Pi}^0_{\eta_m}(X)$, $\xi_n, \eta_m < \xi$. Hence $A_n, B_m \in \mathbf{\Pi}^0_\eta(X)$, $\eta = \max(\xi_n, \eta_m) < \xi$, since $\mathbf{\Pi}^0_\zeta(X) \subseteq \mathbf{\Pi}^0_\eta(X)$ for $\zeta < \eta$ (Proposition 2.8(a)), and $A_n \cap B_m \in \mathbf{\Pi}^0_\eta(X)$ by (a). Thus $\bigcup_{n,m\in\mathbb{N}}(A_n \cap B_m) \in \mathbf{\Sigma}^0_\xi(X)$ by definition. □

Source manuscript page 10

Example 2.10

Recall we proved for X a compact metric space, Y a separable complete metric space, that $C(X,Y)$ is Polish with respect to the uniform metric (HW 1). We show the set of continuously differentiable functions on $[0,1]$, $C^1([0,1])$, is $\mathbf{\Pi}^0_3$ as a subset of $C([0,1]) = C([0,1],\mathbb{R})$, i.e. $C^1([0,1]) \in \mathbf{\Pi}^0_3(C([0,1]))$, i.e. having a definition of the form $\forall^{\mathbb{N}}\exists^{\mathbb{N}}\forall^{\mathbb{N}}$.

We would reduce the complexity of the definition by reducing the complexity of quantifiers (e.g. $\forall \varepsilon \in \mathbb{R}^+ \rightsquigarrow \forall \varepsilon \in \mathbb{Q}^+$) and applying theorems.

For $f \in C([0,1])$, $f \in C^1([0,1]) \iff \forall \varepsilon \in \mathbb{Q}^+$, $\exists$ finitely many rational open intervals $I_0, \dots, I_{n-1}$ covering $[0,1]$ such that $\forall\, 0 \le j < n$, $f \in A_{I_j,\varepsilon}$, where

$$A_{I_j,\varepsilon} := \left\{ f \in C([0,1]) : \forall\, a \neq b,\ c \neq d \text{ in } I_j,\ \left| \frac{f(a)-f(b)}{a-b} - \frac{f(c)-f(d)}{c-d} \right| \le \varepsilon \right\}. \qquad (*)$$

[Exercise. Hint: Recall that $g(x)$ is continuous at x_0 iff for all $x_n \xrightarrow{n\to\infty} x_0$, $g(x_n) \to g(x_0)$, and noticing that $(*)$ implies Cauchyness and convergence (of the difference quotients) at once, we get differentiability at $x \in [0,1]$. Moreover, $(*)$ implies continuity of f' as well.]

Now fixing I_j with $a, b, c, d \in I_j$, the map

$$F_{a,b,c,d} : C([0,1]) \to [0,\infty), \qquad f \mapsto \left| \frac{f(a)-f(b)}{a-b} - \frac{f(c)-f(d)}{c-d} \right|$$

Source manuscript page 11

is easily continuous since every $f \in C([0,1])$ is uniformly continuous. Then

$$A_{I_j,\varepsilon} = \underbrace{\bigcap_{\substack{a,b,c,d\in I_j \\ a\neq b,\ c\neq d}} \underbrace{F^{-1}_{a,b,c,d}([0,\varepsilon])}_{\text{continuous preimage of closed sets = closed}}}_{\bigcap \text{ closed=closed}}$$

is closed in $C([0,1])$. Thus

$$f \in C^1([0,1]) \iff \underbrace{\underbrace{\forall \varepsilon \in \mathbb{Q}^+,\ \exists\, (I_0,\dots,I_{n-1}) \in \text{rational open intervals}^{<\mathbb{N}},\ \underbrace{\forall j < n,\ f \in A_{I_j,\varepsilon}}_{\mathbf{\Pi}^0_1=\text{closed}}}_{\mathbf{\Sigma}^0_2=F_\sigma}}_{\mathbf{\Pi}^0_3=\neg\mathbf{\Sigma}^0_3}$$

(Alternatively, $\mathbf{\Pi}^0_3$ is closed under countable intersections and $\mathbf{\Sigma}^0_2 \subseteq \mathbf{\Delta}^0_3 \subseteq \mathbf{\Pi}^0_3$.) Explicitly, if $\mathcal{I} := \{(p,q) \cap [0,1] : p < q \in \mathbb{Q}\}$, which is countable (! $\mathbb{N}^{<\mathbb{N}}$ is countable, $\mathbb{N}^{\mathbb{N}}$ is uncountable, $= |2^{\mathbb{N}}| = \mathfrak{c}$), then

$$C^1([0,1]) = \bigcap_{\varepsilon\in\mathbb{Q}^+} \bigcup_{\substack{(I_0,I_1,\dots,I_{n-1})\in\mathcal{I}^{<\mathbb{N}} \\ \bigcup_j I_j = [0,1]}} \bigcap_{0\le j<n} A_{I_j,\varepsilon}. \quad \square$$

2.3 The Classes $\mathcal{B}$, $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$

Definition 2.11

By $\mathcal{B}$, $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$, we denote the class of the corresponding sets in all topological spaces, and for each such space X, we write $\mathcal{B}(X)$, $\mathbf{\Sigma}^0_\xi(X)$, $\mathbf{\Pi}^0_\xi(X)$, $\mathbf{\Delta}^0_\xi(X)$ to mean the corresponding set of subsets of X. □

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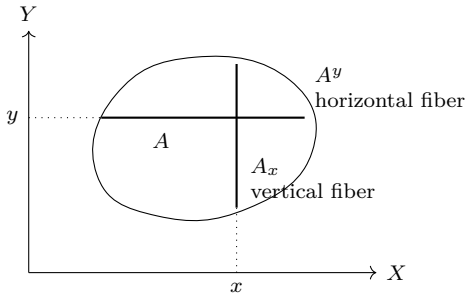
Notation. For a set $A \subseteq X \times Y$, we denote by

$$A_x := \{y \in Y : (x,y) \in A\}, \quad \text{for } x \in X, \qquad \text{and} \qquad A^y := \{x \in X : (x,y) \in A\}, \quad \text{for } y \in Y,$$

the vertical and horizontal fibers over the point $x \in X$ and $y \in Y$ respectively. Similarly, for a function $f : X \times Y \to Z$, denote by

$$f_x : Y \to Z,\ y \mapsto f(x,y) \qquad \text{and} \qquad f^y : X \to Z,\ x \mapsto f(x,y)$$

the vertical and horizontal fibers over the point $x \in X$ and $y \in Y$, respectively. □



Proposition 2.12

Let Γ be any one of the classes $\mathbf{\Sigma}^0_\xi$, $\mathbf{\Pi}^0_\xi$, $\mathbf{\Delta}^0_\xi$, $\xi < \omega_1$. Then Γ is closed under

(a) continuous preimages, i.e. if X, Y are topological spaces, $f : X \to Y$ is continuous and $B \in \Gamma(Y)$, then $f^{-1}(B) \in \Gamma(X)$;

(b) putting countably many sets as fibers over $\mathbb{N}$, i.e. if X is a topological space, $A_n \in \Gamma(X)$, then the set $B \subseteq \mathbb{N} \times X$ defined by $B_n := A_n$ is in $\Gamma(\mathbb{N} \times X)$, where B_n is the fiber of B over $n \in \mathbb{N}$, $\mathbb{N} \times X$ is equipped with the product topology, and $\mathbb{N}$ is equipped with the discrete topology.

Proof. (a) Note that the statement with $\mathbf{\Delta}^0_\xi$ follows from $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ from the fact that taking intersections commutes with taking preimages. We show the statement for $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ simultaneously by transfinite induction on ξ.

Base case: $\xi = 1$. Continuous preimages of open sets ($\mathbf{\Sigma}^0_1(Y)$) are

Source manuscript page 13

open ($\mathbf{\Sigma}^0_1(X)$). $\mathbf{\Pi}^0_1$ follows by taking the dual collection.

Inductive step: Our definition of $\mathbf{\Sigma}^0_\xi$ and $\mathbf{\Pi}^0_\xi$ does not split into successor / limit cases, thus we need not split into cases in our transfinite induction proof. Suppose $\mathbf{\Sigma}^0_\eta$ and $\mathbf{\Pi}^0_\eta$ are closed under continuous preimages, for all $\eta < \xi$. Then for $\bigcup_n A_n \in \mathbf{\Sigma}^0_\xi(Y)$, with $A_n \in \mathbf{\Pi}^0_{\xi_n}(Y)$, $\xi_n < \xi$, $\forall n$, we have $f^{-1}(\bigcup_n A_n) = \bigcup_n f^{-1}(A_n)$ with each $f^{-1}(A_n) \in \mathbf{\Pi}^0_{\xi_n}(X)$, $\xi_n < \xi$, by the induction hypothesis. Thus by definition, $f^{-1}(\bigcup_n A_n) = \bigcup_n f^{-1}(A_n) \in \mathbf{\Sigma}^0_\xi(X)$. (And $\mathbf{\Pi}^0_\xi$ follows by taking the dual collection.)

(b) Induct transfinitely on ξ in the same way as above. Alternatively, consider $\pi : \mathbb{N} \times X \to X$ the projection map, which is continuous by definition of the product topology. Then for $\bigcup_n A_n \in \mathbf{\Sigma}^0_\xi(X)$,

$$B = \bigcup_n (\{n\} \times B_n) = \bigcup_n \left[(\{n\} \times X) \cap \pi^{-1}(A_n) \right]$$

is a countable union of (clopen ($\mathbf{\Delta}^0_1(X) \subseteq \mathbf{\Sigma}^0_\xi(X)$) sets) $\cap$ ($\mathbf{\Sigma}^0_\xi(\mathbb{N} \times X)$ sets). We show that (clopen set) $\cap$ ($\mathbf{\Sigma}^0_\xi$ set) is $\mathbf{\Sigma}^0_\xi$. Indeed, clopen sets are $\mathbf{\Delta}^0_1$, hence $\mathbf{\Pi}^0_1$ and $\mathbf{\Sigma}^0_1$, hence $\mathbf{\Pi}^0_2$ since $\mathbf{\Pi}^0_1 \subseteq \mathbf{\Sigma}^0_2 \implies \mathbf{\Sigma}^0_1 = \neg\mathbf{\Pi}^0_1 \subseteq \neg\mathbf{\Sigma}^0_2 = \mathbf{\Pi}^0_2$, hence $\mathbf{\Pi}^0_\eta \; \forall \eta < \omega_1$ since $\mathbf{\Pi}^0_2 \subseteq \mathbf{\Pi}^0_3 \subseteq \cdots$ in any topological space. Let C be clopen and $\bigcup_n D_n \in \mathbf{\Sigma}^0_\xi$ with $D_n \in \mathbf{\Pi}^0_{\xi_n}$, $\xi_n < \xi \; \forall n$. Then $C \cap (\bigcup_n D_n) = \bigcup_n (C \cap D_n)$, where each $C \cap D_n$ is an intersection in $\mathbf{\Pi}^0_{\xi_n}$. Hence $C \cap D_n \in \mathbf{\Pi}^0_{\xi_n}$ since the $\mathbf{\Pi}^0_\eta$ classes are closed under countable intersections. Thus $\bigcup_n (C \cap D_n) \in \mathbf{\Sigma}^0_\xi$. Hence $B = \bigcup_n \left[(\{n\} \times X) \cap \pi^{-1}(A_n) \right] \in \mathbf{\Sigma}^0_\xi(\mathbb{N} \times X)$.

Source manuscript page 14

For $\mathbf{\Pi}^0_\xi$, notice that $(m, x) \in B = \bigcup_n \{n\} \times B_n \iff \forall n \in \mathbb{N}$, $n \neq m$ or $x \in A_n$, thus

$$B = \underbrace{\bigcap_{n \in \mathbb{N}} \underbrace{\left[\underbrace{((\mathbb{N} \setminus \{n\}) \times X)}_{\mathbf{\Delta}^0_1 \subseteq \mathbf{\Pi}^0_\xi} \cup \underbrace{\pi^{-1}(A_n)}_{\mathbf{\Pi}^0_\xi} \right]}_{\mathbf{\Pi}^0_\xi \; (\dagger)}}_{\mathbf{\Pi}^0_\xi \text{ (by the closure property)}}.$$

To show (†), first show that every set in $\mathbf{\Pi}^0_\xi$ takes the form $\bigcap_n E_n$ with $E_n \in \mathbf{\Sigma}^0_{\xi_n}$, $\xi_n < \xi$, then rewrite the union term as some countable $\bigcap$ in $\mathbf{\Pi}^0_\xi$ (distribute the clopen set into the intersection; a clopen set union a $\mathbf{\Sigma}^0_{\xi_n}$ set is $\mathbf{\Sigma}^0_{\xi_n}$ by the dual of what we just showed). □

Definition 2.13

Let Γ be one of the classes. Denote the dual class by $\neg\Gamma$, i.e. if X is a topological space, then

$$\neg\Gamma(X) := \{A \subseteq X : A^c \in \Gamma(X)\}, \qquad \text{thus } \mathbf{\Sigma}^0_\xi = \neg\mathbf{\Pi}^0_\xi.$$

Furthermore, for topological spaces X, Y and $A \subseteq X \times Y$, denote $\mathrm{proj}_1(A) := \{x \in X : \exists\, y \in Y \text{ such that } (x, y) \in A\}$, and put

$$\exists^Y \Gamma(X) := \{\mathrm{proj}_1(A) : A \in \Gamma(X \times Y)\} \subseteq \mathcal{P}(X),$$
$$\forall^Y \Gamma(X) := \neg\exists^Y \neg\Gamma(X) := \neg(\exists^Y(\neg\Gamma)(X)) \subseteq \mathcal{P}(X).$$

I.e.,

$$\begin{aligned}
S \in \exists^Y \Gamma(X) &\iff \exists\, A \in \Gamma(X \times Y)\ \forall x \in X,\ [x \in S \iff \exists\, y \in Y\ (x,y) \in A],\\
S \in \forall^Y \Gamma(X) &\iff S^c \in (\exists^Y(\neg\Gamma))(X)\\
&\iff \exists\, A \in \neg\Gamma(X \times Y)\ \forall x \in X\ [x \in S^c \iff \exists\, y \in Y\ (x,y) \in A]\\
&\iff \exists\, C \in \Gamma(X \times Y)\ \forall x \in X\ [x \in S \iff \forall y \in Y\ (x,y) \notin A] \quad (C = A^c)\\
&\iff \exists\, C \in \Gamma(X \times Y)\ \forall x \in X\ [x \in S \iff \forall y \in Y\ (x,y) \in C].
\end{aligned}$$

Source manuscript page 15

By $\neg\exists^Y\neg\Gamma(X)$, we mean: apply the operator $\exists^Y$ to the class $\neg\Gamma$ first, then apply $\exists^Y\neg\Gamma$ to X, and finally take the dual collection $\neg(\exists^Y\neg\Gamma(X))$. □

Corollary 2.14

$\forall\, 1 \le \xi < \omega_1$, for X metrizable, $\mathbf{\Sigma}^0_{\xi+1}(X) = \exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$. Thus also $\mathbf{\Pi}^0_{\xi+1}(X) = \forall^{\mathbb{N}}\mathbf{\Sigma}^0_\xi(X)$.

Proof. $\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X) \subseteq \mathbf{\Sigma}^0_{\xi+1}(X)$: Take $S \in \exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$, with $A \in \mathbf{\Pi}^0_\xi(\mathbb{N} \times X)$ such that $\mathrm{proj}_X(A) = S$. Then observe that $\mathrm{proj}_X(A) = \bigcup_{n\in\mathbb{N}} A_n$, with each fiber A_n = the continuous preimage of A by $\iota_n : x \mapsto (n,x)$. By Proposition 2.12(a), each A_n is $\mathbf{\Pi}^0_\xi(X)$. Hence $S \in \mathbf{\Sigma}^0_{\xi+1}(X)$.

$\mathbf{\Sigma}^0_{\xi+1}(X) \subseteq \exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$: Take $S = \bigcup_{n\in\mathbb{N}} A_n$, with $A_n \in \mathbf{\Pi}^0_{\xi_n}(X)$, $\xi_n < \xi + 1$, i.e. $\xi_n \le \xi$. Thus $A_n \in \mathbf{\Pi}^0_\xi(X)$, $\forall n$ (using metrizability so that $\mathbf{\Pi}^0_1 \subseteq \mathbf{\Pi}^0_2$). By Proposition 2.12(b), A with the A_n as fibers over n is $\mathbf{\Pi}^0_\xi(\mathbb{N} \times X)$. Thus $S = \mathrm{proj}_X(A)$ is $\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X)$ by definition.

Lastly $\mathbf{\Pi}^0_{\xi+1}(X) = \neg\mathbf{\Sigma}^0_{\xi+1}(X) = \neg\exists^{\mathbb{N}}\mathbf{\Pi}^0_\xi(X) = \neg\exists^{\mathbb{N}}\neg\mathbf{\Sigma}^0_\xi(X) = \forall^{\mathbb{N}}\mathbf{\Sigma}^0_\xi(X)$. □

Remark 2.15

Thus $\bigcup_{1\le n<\omega} \mathbf{\Sigma}^0_n$ are all the sets obtained from open sets by $\neg$ and $\exists^{\mathbb{N}}$ (finite $\cap$ and $\cup$ can be encoded by $\neg$ and $\exists^{\mathbb{N}}$; in contrast, infinite $\cap$ and $\cup$ could escape to $\mathbf{\Sigma}^0_\omega$ outside of $\bigcup_{1\le n<\omega} \mathbf{\Sigma}^0_n$).

Source manuscript page 16

Because of this, the superscript "0" refers to the fact that we quantify over $\mathbb{N}$ as opposed to $2^{\mathbb{N}}$, i.e. $\mathcal{P}(\mathbb{N})$. $\mathbf{\Sigma}^1_n$, $\mathbf{\Pi}^1_n$ would then be used when we allow quantification over $2^{\mathbb{N}}$ or $\mathbb{N}^{\mathbb{N}}$ or $\mathbb{R}$ or any uncountable Polish space. □

2.4 Universal Sets and Diagonalization

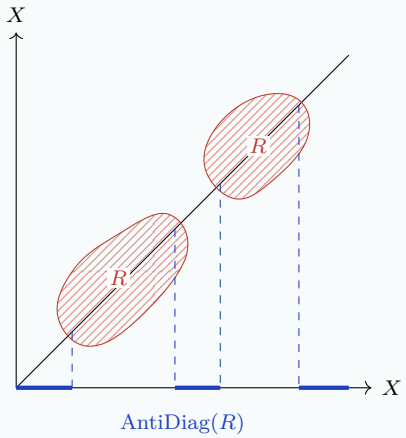
Aside. A very recent work done by Evan Leach (MIT, UCLA, Sep. 2026) presented a combinatorial proof using infinite circuits of what we are going to prove next using diagonalization (the second corollary below, Corollary 2.23).

Lemma 2.16 (Diagonalization Lemma (Cantor))

For a set X and $R \subseteq X \times X$, the antidiagonal set

$$\text{AntiDiag}(R) := \{x \in X : (x, x) \notin R\}$$

is not a vertical or horizontal fiber of R, i.e. $\text{AntiDiag}(R) \neq R_x$ and $\text{AntiDiag}(R) \neq R^x$, $\forall x \in X$.



Proof. Notice that for every $x \in X$, $x \in \text{AntiDiag}(R) \iff (x, x) \notin R \iff x \notin R_x$ (and likewise $\iff x \notin R^x$), thus $\text{AntiDiag}(R) \neq R_x$ (and $\neq R^x$), for any $x \in X$. □

Corollary 2.17

$\mathbb{R}$ and $2^{\mathbb{N}}$ are uncountable.

We illustrate that the lemma above is really generalizing

Source manuscript page 17

the elementary digit-flipping proof of $\mathbb{R}$'s uncountability.

Proof. Suppose $(x_n)_{n\in\mathbb{N}} \subseteq 2^{\mathbb{N}}$, i.e. each x_i is a 0-1 sequence. We can view it as $R \subseteq \mathbb{N} \times \mathbb{N}$ where $(m, n) \in R \iff x_m(n) = 1$. Hence the fiber $R_m = \{n : x_m(n) = 1\}$ is just the set of entries of the m-th sequence having a "1". Consider $\text{AntiDiag}(R) = \{n : x_n(n) = 0\}$. Flipping the digits, we get $y(n) := 1 - x_n(n)$ (the characteristic function of $\text{AntiDiag}(R)$). By Cantor's lemma, $\text{AntiDiag}(R) \neq R_m$, $\forall m$, thus $y \neq x_m$ for any m. So $y \in 2^{\mathbb{N}} \setminus (x_n)_{n\in\mathbb{N}}$. One can inject $2^{\mathbb{N}} \hookrightarrow \mathbb{R}$ via the decimal representation of reals, hence $|\mathbb{N}| < |2^{\mathbb{N}}| \leq |\mathbb{R}|$. □

Exercise 2.18

Use the lemma and show $|X| < |\mathcal{P}(X)|$, i.e. no surjection $X \twoheadrightarrow \mathcal{P}(X)$ exists. (Hint: Suppose such $X \twoheadrightarrow \mathcal{P}(X)$ exists, then view it as a correspondence / subset of $X \times X$. What are the vertical fibres and what is the antidiagonal in this case? Correspondence = set-valued function.) □

Definition 2.19

Let Γ be a class of sets in topological spaces. Let X, Y be topological spaces. We say that a set $U \subseteq Y \times X$ parametrizes $\Gamma(X)$ if $\{U_y : y \in Y\} = \Gamma(X)$. Think of Y as a parameter set, and decoding along Y gives $\Gamma(X)$. If moreover U itself is in Γ, i.e. $U \in \Gamma(Y \times X)$, then we call U a Y-universal set for $\Gamma(X)$. □

Source manuscript page 18

Observation. Let Γ be any class of sets in topological spaces (i.e. Γ is an arbitrary class we can define over a topological space) which is closed under taking continuous preimages. Then Γ is closed under taking fibers, since if $R \in \Gamma(X \times Y)$ and $x \in X$, then R_x is the preimage of R under the continuous function $Y \to X \times Y$, $y \mapsto (x, y)$.

Note 2.20

We have used this observation once already, when we proved Corollary 2.14. □

Theorem 2.21

Let X be second countable metrizable. Then for $1 \leq \xi < \omega_1$, there is a $2^{\mathbb{N}}$-universal set for $\mathbf{\Sigma}^0_\xi(X)$ and for $\mathbf{\Pi}^0_\xi(X)$.

Proof. Transfinite induction on ξ.

$\xi = 1$. Fix a countable basis $(W_n)_{n\in\mathbb{N}}$ for X. Then define $U \subseteq 2^{\mathbb{N}} \times X$: for each $z \in 2^{\mathbb{N}}$,

$$U_z := \bigcup \{W_n : n \in \mathbb{N} \text{ and } z(n) = 1\}.$$

It is clear that $U = \bigcup_{z\in 2^{\mathbb{N}}} \{z\} \times U_z$ parametrizes $\mathbf{\Sigma}^0_1(X)$. Furthermore, U is $\mathbf{\Sigma}^0_1(2^{\mathbb{N}} \times X)$, i.e. open, because

$$\begin{aligned}(z, x) \in U &\iff x \in \bigcup \{W_n : n \in \mathbb{N},\ z(n) = 1\} \\ &\iff \underbrace{\underbrace{\underbrace{\exists n \in \mathbb{N}}_{\text{(countable) union}} \Big(\underbrace{\underbrace{z(n) = 1}_{\substack{\text{clopen} \\ (\{z : z(n)=1\} \times X) \\ \text{clopen (Part 1, trees)}}} \wedge \underbrace{x \in W_n}_{\substack{\text{open} \\ (2^{\mathbb{N}} \times W_n)}}}_{\text{open}} \Big)}_{\text{open}}\end{aligned}$$

(the cylinder sets $\{z : z(n) = 1\}$ are clopen by the note on cylinder sets in Part 1).

Inductive step. Suppose there is a $2^{\mathbb{N}}$-universal set $U^{(\eta)}$ for $\mathbf{\Sigma}^0_\eta(X)$, for each

Source manuscript page 19

$\eta < \xi$. Then $V^{(\eta)} := \left(U^{(\eta)}\right)^c$ is a $2^{\mathbb{N}}$-universal set for $\mathbf{\Pi}^0_\eta(X)$ because fibers commute with complements from the definition (exercise). It remains to build a universal set for $\mathbf{\Sigma}^0_\xi(X)$.

Fix $\eta_n < \xi$ with $\eta_n \leq \eta_{n+1}$, $\forall n \in \mathbb{N}$, such that $\sup\{\eta_n + 1 : n \in \mathbb{N}\} = \xi$. Take $B = \bigcup_k C_k \in \mathbf{\Sigma}^0_\xi(X)$, with each $C_k \in \mathbf{\Pi}^0_{\zeta_k}$, $\zeta_k < \xi$, $\forall k$. We rewrite B as follows: for each $\zeta < \xi = \sup_n(\eta_n + 1)$, there must be

some N with $\zeta \in \eta_N + 1$, i.e. $\zeta < \eta_N + 1$, i.e. $\zeta \leq \eta_N$. Then since the sequence is increasing, $\forall n \geq N$, $\eta_n \geq \eta_N \geq \zeta$. Now give each C_k its own slot: for each k, denote the N for ζ_k by N_k. Inductively set $n_k := \max(N_k, n_{k-1} + 1)$ so that $n_0 < n_1 < n_2 < \cdots$ and $\eta_{n_k} \geq \eta_{N_k} \geq \zeta_k$ from our choice of N. Put $B_{n_k} := C_k$, and for the rest, the unused slots, $B_n := \varnothing$. Observe that we rewrote $B = \bigcup_n B_n$ with $B_n \in \mathbf{\Pi}^0_{\eta_n}$, and particularly $B_{n_k} = C_k \in \mathbf{\Pi}^0_{\zeta_k} \subseteq \mathbf{\Pi}^0_{\eta_{n_k}}$.

Now to encode B, we need to encode all B_n into one code. We use $2^{\mathbb{N}} \cong (2^{\mathbb{N}})^{\mathbb{N}}$ (see the proof of Part 1, Theorem 4.12), so it's enough to show the existence of a $(2^{\mathbb{N}})^{\mathbb{N}}$-universal set for $\mathbf{\Sigma}^0_\xi(X)$. To this end, define $U \subseteq (2^{\mathbb{N}})^{\mathbb{N}} \times X$ by setting, for each $z \in (2^{\mathbb{N}})^{\mathbb{N}}$,

$$U_z := \bigcup_{n \in \mathbb{N}} \left(V^{(\eta_n)}\right)_{z(n)}.$$

From the definition, $U = \bigcup_{z \in (2^{\mathbb{N}})^{\mathbb{N}}} \{z\} \times U_z$ parametrizes $\mathbf{\Sigma}^0_\xi(X)$. Furthermore, U is itself

Source manuscript page 20

in $\mathbf{\Sigma}^0_\xi((2^{\mathbb{N}})^{\mathbb{N}} \times X)$ since

$$(z,x) \in U \iff x \in \bigcup_{n \in \mathbb{N}} \left(V^{(\eta_n)}\right)_{z(n)} \iff \underbrace{\underbrace{\exists n \in \mathbb{N}}_{\text{countable } \cup}\,,\ \underbrace{(z(n),x) \in V^{(\eta_n)}}_{\mathbf{\Pi}^0_{\eta_n}}}_{\mathbf{\Sigma}^0_\xi}$$

(the set of (z,x) with $(z(n),x) \in V^{(\eta_n)}$ is the preimage of $V^{(\eta_n)}$ under the continuous map $(z,x) \mapsto (z(n),x)$, hence $\mathbf{\Pi}^0_{\eta_n}$ by Proposition 2.12(a)). Universality follows. □

Corollary 2.22

Let X be second countable metrizable, Y an uncountable Polish space. For each $1 \leq \xi < \omega_1$, there is a Y-universal set for $\mathbf{\Sigma}^0_\xi(X)$, as well as one for $\mathbf{\Pi}^0_\xi(X)$.

Proof. We build Y-universal sets for $\mathbf{\Pi}^0_\xi(X)$, then take complements to get Y-universal sets for $\mathbf{\Sigma}^0_\xi(X)$. By the perfect set property for Polish spaces (i.e. the Cantor–Bendixson theorem, Part 1, Theorems 5.6 and 5.5), since Y is uncountable, there is a homeomorphic copy $C \subseteq Y$ of $2^{\mathbb{N}}$. Thus by Theorem 2.21, there is a C-universal set $U \in \mathbf{\Pi}^0_\xi(C \times X)$ for $\mathbf{\Pi}^0_\xi(X)$. Clearly U parametrizes $\mathbf{\Pi}^0_\xi(X)$; it remains to observe

$$\mathbf{\Pi}^0_\xi(C \times X) = \mathbf{\Pi}^0_\xi(Y \times X)\big|_{C \times X} := \{A \cap (C \times X) : A \in \mathbf{\Pi}^0_\xi(Y \times X)\}$$

(HW 3, Q2). But because C is closed, for any $A \in \mathbf{\Pi}^0_\xi(Y \times X)$, $A \cap (C \times X) \in \mathbf{\Pi}^0_\xi(Y \times X)$, since $C \times X$ is closed ($\mathbf{\Pi}^0_1(Y \times X)$), thus $\mathbf{\Pi}^0_\xi(Y \times X)$, and $\mathbf{\Pi}^0_\xi$ is closed under countable $\cap$. Hence $U \in \mathbf{\Pi}^0_\xi(C \times X) = \mathbf{\Pi}^0_\xi(Y \times X)|_{C \times X} \subseteq \mathbf{\Pi}^0_\xi(Y \times X)$, so U is Y-universal for $\mathbf{\Pi}^0_\xi(X)$. □

Corollary 2.23

For every uncountable Polish space X and every $1 \leq \xi < \omega_1$, $\mathbf{\Sigma}^0_\xi(X) \neq \mathbf{\Pi}^0_\xi(X)$. Consequently, we get the so-called strictness of the Borel hierarchy in uncountable Polish spaces,

Source manuscript page 21

that

$$\mathbf{\Delta}^0_1(X) \subsetneq \begin{matrix}\mathbf{\Sigma}^0_1(X)\\ \mathbf{\Pi}^0_1(X)\end{matrix} \subsetneq \mathbf{\Delta}^0_2(X) \subsetneq \begin{matrix}\mathbf{\Sigma}^0_2(X)\\ \mathbf{\Pi}^0_2(X)\end{matrix} \subsetneq \cdots$$

Proof. Let U be an X-universal set for $\mathbf{\Sigma}^0_\xi(X)$, and $A := \mathrm{AntiDiag}(U)$. Then by definition $A = \delta^{-1}(U^c)$ with $\delta : X \to X \times X$ given by $\delta(x) = (x, x)$. Moreover, $A \in \mathbf{\Pi}^0_\xi(X)$ since $U^c \in \mathbf{\Pi}^0_\xi(X \times X)$, δ is continuous, and $\mathbf{\Pi}^0_\xi$ is closed under continuous preimages (Proposition 2.12(a)). But A is not in $\mathbf{\Sigma}^0_\xi(X)$ because A is not a fiber of U (Lemma 2.16), yet U parametrizes $\mathbf{\Sigma}^0_\xi(X)$. □

Source manuscript page 22

3 Change of Topology

3.1 Turning Borel Sets into Clopen Sets

Notation. To be precise, we denote a Polish space X by $(X, \mathcal{T})$, where $\mathcal{T}$ is the topology of X. We have the following most useful tool with Borel sets.

Theorem 3.1

Let $(X, \mathcal{T})$ be a Polish space. For each Borel set $A \subseteq X$, there is a finer Polish topology $\mathcal{T}_A$ with respect to which A is clopen and $\mathcal{B}(\mathcal{T}_A) = \mathcal{B}(\mathcal{T})$.

We prove two lemmas first.

Lemma 3.2

Let $(X, \mathcal{T})$ be a Polish space, $F \subseteq X$ a closed set. Then the topology $\mathcal{T}_F$ generated by $\mathcal{T} \cup \{F\}$ is Polish, with F clopen in $\mathcal{T}_F$ and $\mathcal{B}(\mathcal{T}_F) = \mathcal{B}(\mathcal{T})$.

Proof. We just need to show $\mathcal{T}_F$ is Polish. Observe that $\mathcal{T}_F$ is nothing but the disjoint union topology of F and $X \setminus F$ with their relative topologies respectively, both of which are Polish since F and $X \setminus F$ are both G_δ subsets of the Polish space $(X, \mathcal{T})$ (Part 1, Theorem 2.9), i.e. the map

F	$X \setminus F$

$$(X, \mathcal{T}_F) \to \{0\} \times F \sqcup \{1\} \times (X \setminus F), \qquad x \mapsto \begin{cases}(0, x) & \text{if } x \in F;\\ (1, x) & \text{otherwise}\end{cases}$$

is (trivially) a homeomorphism. But we showed on day 1 (Part 1, Proposition 2.4(b)) that a disjoint union of Polish spaces is Polish (with respect to $d((n, x), (m, y)) := d_n(x, y)$ if $n = m$; 1 otherwise). Hence $(X, \mathcal{T}_F)$

Source manuscript page 23

is Polish. In particular, $\mathcal{B}(X, \mathcal{T}_F) = \mathcal{B}(X, \mathcal{T})$ since F is closed, hence Borel in $\mathcal{T}$. □

Lemma 3.3

Let $(X, \mathcal{T})$ be a Polish space, $(\mathcal{T}_n)_{n\in\mathbb{N}}$ a sequence of Polish topologies on X refining $\mathcal{T}$ with $\mathcal{T}_n \subseteq \mathcal{B}(\mathcal{T})$ (i.e. $\mathcal{B}(\mathcal{T}_n) = \mathcal{B}(\mathcal{T})$). Then the topology $\mathcal{T}_\infty$ generated by $\bigcup_{n\in\mathbb{N}} \mathcal{T}_n$ is Polish with $\mathcal{B}(\mathcal{T}_\infty) = \mathcal{B}(\mathcal{T})$.

Proof. Take a countable basis $\mathcal{U}_n$ for each $\mathcal{T}_n$. Then $\mathcal{T}_\infty$ is generated by the finite intersections of sets from $\bigcup_n \mathcal{U}_n$. So every open set in $\mathcal{T}_\infty$ is some countable union (by second countability of $\mathcal{T}_\infty$) of finite intersections of sets from $\bigcup_n \mathcal{U}_n$, hence $\mathcal{B}(\mathcal{T})$. Thus $\mathcal{T}_\infty \subseteq \mathcal{B}(\mathcal{T}) \implies \mathcal{B}(\mathcal{T}_\infty) \subseteq \mathcal{B}(\mathcal{T})$ since $\mathcal{B}(\mathcal{T}_\infty)$ is the smallest σ-algebra containing $\mathcal{T}_\infty$ (and $\supseteq$ holds since $\mathcal{T} \subseteq \mathcal{T}_\infty$).

It is left to show that $(X, \mathcal{T}_\infty)$ admits a complete compatible metric. We show that $(X, \mathcal{T}_\infty)$ is homeomorphic to the diagonal $\Delta := \{(x)_{n\in\mathbb{N}} : x \in X\} \subseteq X^{\mathbb{N}}$ equipped with the product topology, via

$$\delta : (X, \mathcal{T}_\infty) \to \Big(X^{\mathbb{N}}, \prod_{n\in\mathbb{N}} \mathcal{T}_n\Big), \qquad x \mapsto (x)_n = (x, x, \dots).$$

Clearly δ is a bijection onto Δ. δ is continuous since each coordinate map $\delta_n := \mathrm{proj}_n \circ \delta : (X, \mathcal{T}_\infty) \to (X, \mathcal{T}_n)$ is continuous: δ_n is simply the identity map from a finer topology to a coarser topology, hence continuous. Then it is easy to check that the coordinate maps δ_n being

Source manuscript page 24

continuous $\implies$ δ continuous. On the other hand, to show $\delta^{-1} : \Delta \to (X, \mathcal{T}_\infty)$ is continuous, it suffices to show that preimages of open sets in $\mathcal{T}_n$ by δ^{-1} are open, since $\mathcal{T}_\infty$ is generated by $\bigcup_n \mathcal{T}_n$, and preimages commute with $\cap$ and $\cup$. But for any $n \in \mathbb{N}$, $\delta^{-1} = \mathrm{proj}_n|_\Delta$, hence for $U \in \mathcal{T}_n$, $(\delta^{-1})^{-1}(U) = (\mathrm{proj}_n|_\Delta)^{-1}(U)$ is open in the product topology since the proj_n are continuous by definition of the product topology.

Therefore $(X, \mathcal{T}_\infty) \xrightarrow{\delta} \Delta \subseteq X^{\mathbb{N}}$ is a homeomorphism. Furthermore, Δ is closed since each $\mathcal{T}_n$ is Hausdorff (exercise: show the complement is open). Hence Δ is Polish since it is a G_δ subset of a Polish space (a countable product of Polish spaces is Polish, Part 1, Proposition 2.4(c), and closed sets are G_δ, Part 1, Theorem 2.9). Thus $(X, \mathcal{T}_\infty)$ is Polish. □

Proof of Theorem 3.1 (Clopenification). We do the same σ-algebra trick. Let

$$\mathcal{S} := \{A \in \mathcal{B}(\mathcal{T}) : \exists \text{ Polish } \mathcal{T}_A \supseteq \mathcal{T} \text{ with } A \text{ clopen in } \mathcal{T}_A \text{ and } \mathcal{B}(\mathcal{T}_A) = \mathcal{B}(\mathcal{T})\}.$$

$\mathcal{S} \supseteq$ all closed sets by Lemma 3.2. $\mathcal{S}$ is closed under complements since clopenness is. Finally, for $A_n \in \mathcal{S}$, there are Polish $\mathcal{T}_n \supseteq \mathcal{T}$, $\mathcal{B}(\mathcal{T}_n) = \mathcal{B}(\mathcal{T})$, with A_n clopen in $\mathcal{T}_n$; by Lemma 3.3, $\mathcal{T}_\infty$ is Polish with $\bigcup_n A_n$ open in $\mathcal{T}_\infty$. Lastly apply Lemma 3.2 again (to the closed set $\big(\bigcup_n A_n\big)^c$) so that $\bigcup_n A_n$ is clopen in $\widetilde{\mathcal{T}}_\infty \supseteq \mathcal{T}_\infty$ with $\mathcal{B}(\widetilde{\mathcal{T}}_\infty) = \mathcal{B}(\mathcal{T}_\infty) = \mathcal{B}(\mathcal{T})$. Hence $\bigcup_n A_n \in \mathcal{S}$, i.e. $\mathcal{S}$ is

Source manuscript page 25

closed under countable $\cup$, hence a σ-algebra. But open sets ($= \mathcal{T}$) are in $\mathcal{S}$ (since closed sets are) and $\mathcal{S} \subseteq \mathcal{B}(\mathcal{T}) \implies \langle\mathcal{T}\rangle_\sigma = \mathcal{B}(\mathcal{T}) = \mathcal{S}$. □

Caution. We proved that a closed (thus the same for an open) set can be made clopen by simply taking the topology $\mathcal{T}_F$ generated by $\mathcal{T} \cup \{F\}$. However, it is not true that if B is a Borel set in a Polish space $(X, \mathcal{T})$, then the topology generated by $\mathcal{T} \cup \{B, B^c\}$ is Polish. To obtain a Polish topology in

which B is clopen, one has to clopenify all (countably many) sets that go into the definition of B from open / closed sets (what we did in Lemma 3.3). □

Corollary 3.4

Let $\mathcal{U}$ be a countable collection of Borel sets in a Polish space $(X, \mathcal{T})$. Then there is a Polish topology $\mathcal{T}' \supseteq \mathcal{T}$ in which every set in $\mathcal{U}$ is clopen and $\mathcal{B}(\mathcal{T}') = \mathcal{B}(\mathcal{T})$.

Proof. For each $U \in \mathcal{U}$, let $\mathcal{T}_U \supseteq \mathcal{T}$ be a Polish topology in which U is clopen (Theorem 3.1). Then by Lemma 3.3 above, the topology generated by $\bigcup_{U\in\mathcal{U}} \mathcal{T}_U$ is Polish with the desired property. □

Definition 3.5

Recall that a topological space X has the perfect set property (PSP) if X is either countable or there is a nonempty perfect subset (i.e. perfect closed) of X.

Source manuscript page 26

If X is Polish, then X has the PSP iff X is countable or $2^{\mathbb{N}}$ topologically embeds into X: ($\Rightarrow$) is by the perfect set theorem from Part 1, §5 (Part 1, Theorem 5.4). ($\Leftarrow$) since $2^{\mathbb{N}}$ is compact, thus compact in the Hausdorff space X, hence closed in X (Part 1, Theorem 4.7(b)). Thus $2^{\mathbb{N}}$ is a nonempty perfect (closed) subset of X. □

We now give the corollary that not only Polish spaces and Polish subspaces (i.e. G_δ subsets of Polish spaces) have the PSP, but the Borel sets also have it.

Corollary 3.6

Every Borel set B in a Polish space $(X, \mathcal{T})$ has the perfect set property.

Proof. Take a Polish topology $\mathcal{T}' \supseteq \mathcal{T}$ on X in which B is clopen, hence Polish in the subspace topology of $\mathcal{T}'$. By the Cantor–Bendixson theorem (Part 1, Theorem 5.5), B has the PSP, so if B is uncountable, then there is a continuous embedding $f : 2^{\mathbb{N}} \hookrightarrow (B, \mathcal{T}'|_B)$. Then trivially, f is still continuous as a map to $(B, \mathcal{T}|_B)$ and still injective, hence an embedding into $(B, \mathcal{T}|_B)$ because $2^{\mathbb{N}}$ is compact and $\mathcal{T}|_B$ is Hausdorff (Part 1, Theorem 4.7(e)). □

Corollary 3.7 (Making Borel functions continuous)

Let $(X, \mathcal{T})$ be a Polish space and Y be a second countable topological space. Let $f : X \to Y$ be a Borel (measurable) function. Then there is a Polish topology $\mathcal{T}_f \supseteq \mathcal{T}$ such that $f : (X, \mathcal{T}_f) \to Y$ is continuous and $\mathcal{B}(\mathcal{T}_f) = \mathcal{B}(\mathcal{T})$.

Source manuscript page 27

Proof. Let $\mathcal{U}$ be a countable basis for Y, and $\mathcal{T}_f \supseteq \mathcal{T}$ a Polish topology with $\mathcal{B}(\mathcal{T}_f) = \mathcal{B}(\mathcal{T})$ in which every set $f^{-1}(U)$ is clopen, for all $U \in \mathcal{U}$ (Corollary 3.4). Hence f is continuous from $(X, \mathcal{T}_f)$ to Y. □

Corollary 3.8

Let $(X, \mathcal{T})$ be a Polish space, $B \subseteq X$ a Borel subset. Then there is a continuous bijection $f : F \to B$ for some closed $F \subseteq \mathbb{N}^{\mathbb{N}}$. In particular, if $B \neq \varnothing$, then there is a continuous surjection $\mathbb{N}^{\mathbb{N}} \twoheadrightarrow B$.

Proof. Let $\mathcal{T}' \supseteq \mathcal{T}$ be a Polish topology with B clopen. Hence $(B, \mathcal{T}'|_B)$ is Polish, so there is a continuous bijection $f : F \xrightarrow{\sim} (B, \mathcal{T}'|_B)$, where $F \subseteq \mathbb{N}^{\mathbb{N}}$ is closed (Part 1, Theorem 6.14). If $B \neq \varnothing$, then there is a continuous surjection $g : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow (B, \mathcal{T}'|_B)$, composing f with a retraction $\mathbb{N}^{\mathbb{N}} \twoheadrightarrow F$ (Part 1, Proposition 3.17). But then f, g are trivially continuous with respect to the coarser topology $\mathcal{T}|_B$. □

Proposition 3.9

Let $(X, \mathcal{T})$ be a Polish space. Then there exists a zero-dimensional Polish topology $\mathcal{T}_0 \supseteq \mathcal{T}$ such that $\mathcal{B}(\mathcal{T}_0) = \mathcal{B}(\mathcal{T})$.

Proof. HW 3. □

Proposition 3.10

Let Γ be a countable group, and consider a Borel action of Γ on a Polish space $(X, \mathcal{T})$, i.e. every $g \in \Gamma$ acts as a Borel automorphism of X. Then there exists a Polish topology $\mathcal{T}_0 \supseteq \mathcal{T}$ with $\mathcal{B}(\mathcal{T}_0) = \mathcal{B}(\mathcal{T})$ that makes the action of Γ continuous. Moreover, $\mathcal{T}_0$ can be taken to be zero-dimensional.

Source manuscript page 28

Proof. HW 3. □

3.2 Making Closed-Graph Functions Continuous for Free

Definition 3.11

Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be topological spaces, $f : X \to Y$. The topology generated by f over $\mathcal{T}_X$ is the coarsest topology $\mathcal{T}_{X,f} \supseteq \mathcal{T}_X$ which makes f continuous. □

Observation. $\mathcal{T}_{X,f}$ is the topology generated by $\mathcal{T}_X \cup \{f^{-1}(V) : V \in \mathcal{T}_Y\}$. Every $U \in \mathcal{T}_{X,f}$ is an arbitrary union of sets of the form $U \cap f^{-1}(V)$, $U \in \mathcal{T}_X$, $V \in \mathcal{T}_Y$.

Lemma 3.12

Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be topological spaces and $f : X \to Y$. Then we have two topologies on $X \times Y$: $\mathcal{T}_X \times \mathcal{T}_Y$ and $\mathcal{T}_{X,f} \times \mathcal{T}_Y$. On the graph $G := \mathrm{graph}(f) = \{(x, f(x)) : x \in X\} \subseteq X \times Y$, the relative topologies with respect to $\mathcal{T}_X \times \mathcal{T}_Y$ and $\mathcal{T}_{X,f} \times \mathcal{T}_Y$ coincide.

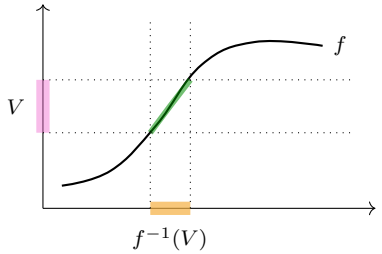
Proof. Notice that the basic open sets in $\mathcal{T}_{X,f} \times \mathcal{T}_Y|_G$ have the form

$$[(U \cap f^{-1}(V)) \times W] \cap G = [(U \times W) \cap G] \cap [(f^{-1}(V) \times W) \cap G].$$

But $U \times W$ is open in $\mathcal{T}_X \times \mathcal{T}_Y$, thus $(U \times W) \cap G$ is open in $\mathcal{T}_X \times \mathcal{T}_Y|_G$. It is left to show that $(f^{-1}(V) \times W) \cap G$ is open in $\mathcal{T}_X \times \mathcal{T}_Y|_G$. But this is simply because

$$(f^{-1}(V) \times W) \cap G = [X \times (V \cap W)] \cap G,$$

which is open in $\mathcal{T}_X \times \mathcal{T}_Y|_G$ since $X \in \mathcal{T}_X$ and $V \cap W \in \mathcal{T}_Y$.



$(f^{-1}(V) \times Y) \cap G = (X \times V) \cap G$

□

Lemma 3.13 (Continuity via graph)

Let X, Y be topological spaces. A

Source manuscript page 29

function $f : X \to Y$ is continuous

$$\iff \pi_f : X \to \text{graph}(f),\ x \mapsto (x, f(x)) \text{ is a homeomorphism} \iff \pi_f \text{ is continuous.}$$

Proof. If π_f is continuous, then so is f since $f = \text{proj}_Y \circ \pi_f$. On the other hand, if f is continuous, then so is π_f since each coordinate function $x \mapsto x$, $x \mapsto f(x)$ is continuous, hence continuity of π_f by definition of the product topology. Lastly, we get π_f^{-1} continuous for free since $\pi_f^{-1} = \text{proj}_X|_{\text{graph}(f)}$. □

Proposition 3.14 (Ruiyuan Chen)

Let $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$ be Polish spaces, $f : X \to Y$. If $\text{graph}(f)$ is closed in $\mathcal{T}_X \times \mathcal{T}_Y$ (for instance, if f is continuous with respect to some coarser Hausdorff topology on Y), then $\mathcal{T}_{X,f}$ is Polish.

Proof. $f : (X, \mathcal{T}_{X,f}) \to Y$ is continuous by definition of $\mathcal{T}_{X,f}$. Set $G := \text{graph}(f)$. Then $\pi_f : (X, \mathcal{T}_{X,f}) \to (G, \mathcal{T}_{X,f} \times \mathcal{T}_Y|_G)$ is a homeomorphism (Lemma 3.13). Then $(G, \mathcal{T}_X \times \mathcal{T}_Y|_G)$ is Polish, being a closed subset of the Polish space $(X \times Y, \mathcal{T}_X \times \mathcal{T}_Y)$, and $\mathcal{T}_X \times \mathcal{T}_Y|_G = \mathcal{T}_{X,f} \times \mathcal{T}_Y|_G$ by Lemma 3.12. Thus $(X, \mathcal{T}_{X,f}) \cong (G, \mathcal{T}_X \times \mathcal{T}_Y|_G) = (G, \mathcal{T}_{X,f} \times \mathcal{T}_Y|_G)$ is Polish. □

Source manuscript page 30

4 Analytic Sets

4.1 Definition and Closure Properties

We know that the class of Borel sets is closed under continuous / Borel preimages. Lebesgue had a false proof that it is closed under continuous images. It was discovered by Souslin years later, who proved that these are new sets and studied them with his PhD advisor Luzin.

Definition 4.1

A subset A of a Polish space X is called analytic if it is a continuous image of a Borel subset of some Polish space, i.e. if there are a Polish space Y, a Borel subset $B \subseteq Y$, and a continuous function $f : Y \to X$ such that $A = f(B)$. □

Proposition 4.2 (Equivalence of analyticity)

Let X be Polish, $\varnothing \neq A \subseteq X$. Then the following are equivalent:

(1) A is analytic (the definition above).
(2) There are a Polish space Y and a continuous $f : Y \to X$ such that $f(Y) = A$.
(3) There is a continuous $f : \mathbb{N}^{\mathbb{N}} \to X$ such that $f(\mathbb{N}^{\mathbb{N}}) = A$.
⋆(4) There is a closed $F \subseteq X \times \mathbb{N}^{\mathbb{N}}$ such that $\mathrm{proj}_X(F) = A$.
(5) There are a Polish Y and a Borel $B \subseteq X \times Y$ such that $\mathrm{proj}_X(B) = A$.
(6) There are a Polish Y, a Borel $B \subseteq Y$ and a Borel (measurable) $f : Y \to X$ such that $f(B) = A$.

Proof. Immediately, (4) ⇒ (5) ⇒ (1) and (1) ⇒ (6). (1) ⇒ (2) follows by a change of topology to make B clopen (Theorem 3.1), hence Polish with a finer topology $\mathcal{T}_B$, but then f is just more continuous. (2) ⇒ (3) since there is a continuous surjection $\mathbb{N}^{\mathbb{N}} \twoheadrightarrow Y$ (Part 1, Theorem 6.14), thus we can

Source manuscript page 31

compose it with f. (3) ⇒ (4) follows by taking

$$F = \mathrm{graph}(f)^{-1} := \{(f(x), x) : x \in \mathbb{N}^{\mathbb{N}}\}$$

since the graph of a continuous function (into a Hausdorff space) is closed, and $A = f(\mathbb{N}^{\mathbb{N}}) = \mathrm{proj}_X(F)$. (6) ⇒ (1) since we can make a Borel measurable function into a continuous function by Corollary 3.7. □

Notation. We denote by $\mathbf{\Sigma}^1_1$ the class of all analytic subsets of Polish spaces, so given a Polish space X, the set of analytic subsets of X is denoted by $\mathbf{\Sigma}^1_1(X)$. Its dual class is denoted by $\mathbf{\Pi}^1_1 := \neg\mathbf{\Sigma}^1_1$, with

$$\mathbf{\Pi}^1_1(X) = \neg\mathbf{\Sigma}^1_1(X) := \{A^c : A \in \mathbf{\Sigma}^1_1(X)\},$$

its elements called coanalytic sets. We also have the ambiguous class $\mathbf{\Delta}^1_1 := \mathbf{\Pi}^1_1 \cap \mathbf{\Sigma}^1_1$.

By (4) above, we have that $\mathbf{\Sigma}^1_1 = \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Pi}^0_1$, i.e. $\mathbf{\Sigma}^1_1(X) = \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Pi}^0_1(X)$, as we discussed in Remark 2.15. Taking its dual, we get $\mathbf{\Pi}^1_1(X) = \forall^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Sigma}^0_1(X)$. This is what will allow us to build a universal set for $\mathbf{\Sigma}^1_1$, hence an analytic set which is not Borel, via Cantor's diagonalization. □

Observation. $\mathcal{B}(X) \subseteq \mathbf{\Delta}^1_1(X)$ since by (6), taking $f = \mathrm{id}$, all Borel sets are analytic; but $\mathcal{B}(X)$ is closed under taking complements, hence they are also coanalytic. □

Proposition 4.3

The class $\mathbf{\Sigma}^1_1$ is closed under

Source manuscript page 32

(i) continuous images and preimages (in fact, Borel images and preimages);
(ii) countable unions and countable intersections.

Proof. (i) Let X, Y be Polish spaces, $g : X \to Y$ continuous, $\varnothing \neq A \subseteq X$ an analytic subset. Then there is a continuous $f : \mathbb{N}^{\mathbb{N}} \to X$ with $f(\mathbb{N}^{\mathbb{N}}) = A$. Thus $g(A) \subseteq Y$ is witnessed by $g \circ f : \mathbb{N}^{\mathbb{N}} \to Y$.

Now suppose $B \subseteq Y$ is analytic. Then there is a closed $F \subseteq Y \times \mathbb{N}^{\mathbb{N}}$ such that $B = \mathrm{proj}_Y(F)$. Define $\tilde{g} : X \times \mathbb{N}^{\mathbb{N}} \to Y \times \mathbb{N}^{\mathbb{N}}$ as $\tilde{g}((x, n)) := (g(x), n)$, which is continuous since its component maps are. Hence $\tilde{g}^{-1}(F)$ is closed. Moreover, observe that

$$\begin{aligned} x \in g^{-1}(B) &\iff g(x) \in B = \mathrm{proj}_Y(F) \iff \exists\, n \in \mathbb{N}^{\mathbb{N}} \text{ such that } (g(x), n) \in F \\ &\iff \exists\, n \in \mathbb{N}^{\mathbb{N}} \text{ such that } (x, n) \in \tilde{g}^{-1}(F) \iff x \in \mathrm{proj}_X\left(\tilde{g}^{-1}(F)\right). \end{aligned}$$

Thus $g^{-1}(B) = \mathrm{proj}_X\left(\tilde{g}^{-1}(F)\right)$, with the latter an analytic set by condition (4) of Proposition 4.2 and the fact that $\tilde{g}^{-1}(F)$ is closed (since $\tilde{g}$ is continuous). Borel functions follow from the fact that they can be made continuous in a finer Polish topology on X (Corollary 3.7).

(ii) Countable intersections: Let X be a Polish space, $A_n \subseteq X$ analytic. Then there are closed $C_n \subseteq X \times \mathbb{N}^{\mathbb{N}}$ such that $A_n = \mathrm{proj}_X(C_n)$, i.e. for each $x \in X$, $x \in A_n \iff \exists\, y \in \mathbb{N}^{\mathbb{N}}$, $(x, y) \in C_n$. Let $A := \bigcap_{n \in \mathbb{N}} A_n$. Then for $x \in X$,

$$x \in A \iff \forall n \in \mathbb{N},\ \exists\, y_n \in \mathbb{N}^{\mathbb{N}},\ (x, y_n) \in C_n \iff \exists\, y \in (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}\ \forall n \in \mathbb{N},\ (x, y(n)) \in C_n.$$

Source manuscript page 33

Hence with $\widetilde{C}_n := \{(x, y) \in X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} : (x, y(n)) \in C_n\}$, we get the following:

(a) $A = \mathrm{proj}_X\left(\bigcap_n \widetilde{C}_n\right)$ (trivial).
(b) The $\widetilde{C}_n$ are closed: Observe that the map $\pi_n : X \times (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}} \to X \times \mathbb{N}^{\mathbb{N}}$, $(x, y) \mapsto (x, y(n))$, is continuous since each component map is a projection, and $\widetilde{C}_n = \pi_n^{-1}(C_n)$ with C_n closed, hence the $\widetilde{C}_n$ are closed.

Thus $\bigcap_n \widetilde{C}_n$ is closed, thus A is analytic by (a) and the fact that $\mathbb{N}^{\mathbb{N}} \cong \mathbb{N}^{\mathbb{N} \times \mathbb{N}} \cong (\mathbb{N}^{\mathbb{N}})^{\mathbb{N}}$. We leave the countable union as an exercise (HW 3). □

4.2 Universal Sets for Analytic / Coanalytic Sets

In the same way we proved the strictness of the Borel hierarchy in uncountable Polish spaces with universal sets and Cantor's diagonalization, we show the analogue for analytic / coanalytic sets, that $\mathbf{\Pi}^1_1 \neq \mathbf{\Sigma}^1_1$.

Theorem 4.4 (Souslin)

For every uncountable Polish Y and an arbitrary Polish X, there is a Y-universal set for $\mathbf{\Sigma}^1_1(X)$. The same holds for $\mathbf{\Pi}^1_1(X)$.

Proof. We use $\mathbf{\Sigma}^1_1 = \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Pi}^0_1$. Let $F \subseteq Y \times (X \times \mathbb{N}^{\mathbb{N}})$ be a Y-universal set for $\mathbf{\Pi}^0_1(X \times \mathbb{N}^{\mathbb{N}})$ (Corollary 2.22). Let $U := \mathrm{proj}_{Y\times X}(F)$. Then we can verify that U is Y-universal for $\mathbf{\Sigma}^1_1(X)$

Source manuscript page 34

because fibers commute with projections:

$$\{U_y : y \in Y\} = \{\mathrm{proj}_X(F_y) : y \in Y\} = \{\mathrm{proj}_X(C) : C \in \mathbf{\Pi}^0_1(X \times \mathbb{N}^{\mathbb{N}})\} = \mathbf{\Sigma}^1_1(X)$$

by definition, and $U = \mathrm{proj}_{Y\times X}(F) \in \mathbf{\Sigma}^1_1(Y \times X)$ by definition as well. □

Corollary 4.5 (Souslin)

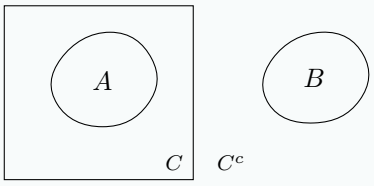
For each uncountable Polish space X, $\mathbf{\Sigma}^1_1(X) \neq \mathbf{\Pi}^1_1(X)$. In particular, $\mathcal{B}(X) \subseteq \mathbf{\Delta}^1_1(X) \subsetneq \mathbf{\Sigma}^1_1(X), \mathbf{\Pi}^1_1(X)$.

Proof. Let $U \subseteq X \times X$ be an X-universal set for $\mathbf{\Sigma}^1_1(X)$. Then $A := \mathrm{AntiDiag}(U) = \{x \in X : (x, x) \notin U\}$ is coanalytic because $A = \delta^{-1}(U^c) = \delta^{-1}(U)^c$, where $\delta : X \to X^2$, $x \mapsto (x, x)$, is continuous and $\mathbf{\Sigma}^1_1$ is closed under continuous preimages. But A is not analytic because $A \neq U_x$, $\forall x \in X$ (Lemma 2.16), while $\{U_x : x \in X\} = \mathbf{\Sigma}^1_1(X)$. □

4.3 Analytic Separation

Theorem 4.6 (Luzin)

Let X be Polish and $A, B \subseteq X$ be disjoint analytic sets. Then A, B are Borel-separable, i.e. there is a Borel $C \subseteq X$ such that $C \supseteq A$ and $C^c \supseteq B$.

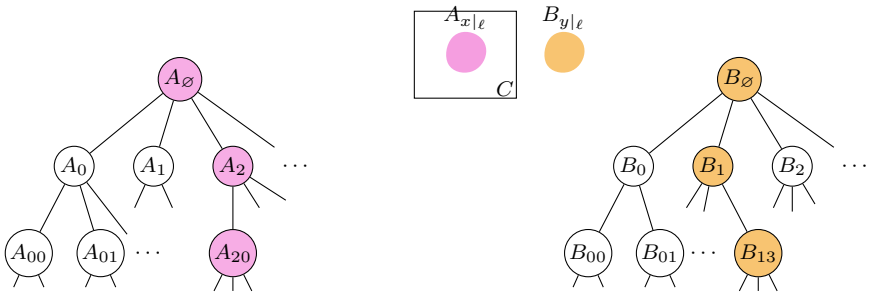


Proof. For a set $P \subseteq X$, observe that the sets which are Borel-separable from P form a σ-ideal, i.e. if each Q_n is Borel-separable from P by a Borel set B_n, then P is Borel-separable from $\bigcup_n Q_n$ by $\bigcup_n B_n$: $\bigcup_n B_n \supseteq \bigcup_n Q_n$ and $(\bigcup_n B_n) \cap P = \bigcup_n (B_n \cap P) = \varnothing$. Hence moreover, if

Source manuscript page 35

P_n, Q_m are Borel-separable for all $m, n \in \mathbb{N}$, then $\bigcup_n P_n$ and $\bigcup_m Q_m$ are Borel-separable: separate P_n from $\bigcup_m Q_m$ for each n, then separate $\bigcup_n P_n$ from $\bigcup_m Q_m$.

Now fix nonempty disjoint analytic sets A, B (if one is empty, then the case is trivial). Then there are continuous surjections $f : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow A$ and $g : \mathbb{N}^{\mathbb{N}} \twoheadrightarrow B$. For $s \in \mathbb{N}^{<\mathbb{N}}$, put $A_s := f([s]_{\mathbb{N}})$, $B_s := g([s]_{\mathbb{N}})$, so $A_s = \bigcup_{n\in\mathbb{N}} A_{sn}$ and $B_s = \bigcup_{n\in\mathbb{N}} B_{sn}$, and $A = A_\varnothing$ and $B = B_\varnothing$. If A, B are not separable, then since $A = A_\varnothing = \bigcup_n A_n$, $B = B_\varnothing = \bigcup_m B_m$, there are $n_0, m_0 \in \mathbb{N}$ such that A_{n_0} and B_{m_0} are not Borel-separable. Further, since $A_s = \bigcup_n A_{sn}$, $B_s = \bigcup_n B_{sn}$, there are $n_1, m_1 \in \mathbb{N}$ such that $A_{n_0 n_1}$ and $B_{m_0 m_1}$ are not Borel-separable. Hence by induction, there are infinite branches $x, y \in \mathbb{N}^{\mathbb{N}}$ such that for every $\ell \in \mathbb{N}$, $A_{x|_\ell}$, $B_{y|_\ell}$ are not Borel-separable.



But $f(x)$ and $g(y)$ are distinct because $f(x) \in A$ and $g(y) \in B$, hence separable by open sets by Hausdorffness, i.e. there are $U, V \subseteq X$ disjoint open such that $U \ni f(x)$ and $V \ni g(y)$. By

Source manuscript page 36

the continuity of f and g and the fact that $([x|_\ell])_\ell$ and $([y|_\ell])_\ell$ form neighbourhood bases at x, y respectively, there is $\ell \in \mathbb{N}$ such that $[x|_\ell] \subseteq f^{-1}(U)$, $[y|_\ell] \subseteq g^{-1}(V)$. But then $A_{x|_\ell} = f([x|_\ell])$ and $B_{y|_\ell} = g([y|_\ell])$ are Borel-separable by either U or V. $\Rightarrow\!\Leftarrow$ □

Corollary 4.7 (Souslin)

Let X be Polish. Then $\mathcal{B}(X) = \mathbf{\Delta}^1_1(X)$.

Proof. It is left to prove $\mathbf{\Delta}^1_1(X) \subseteq \mathcal{B}(X)$ (the other inclusion is the observation above). Take $A \in \mathbf{\Delta}^1_1(X)$. Then both A and A^c are analytic. Hence by analytic separation (Theorem 4.6), there is $B \in \mathcal{B}(X)$ such that $B \supseteq A$, $B^c \supseteq A^c$. Thus $B = A$ and $A \in \mathcal{B}(X)$. □

Remark 4.8

This gives a powerful method for proving that a set is Borel. □

Corollary 4.9

Let X, Y be Polish, $f : X \to Y$. The following are equivalent:

(1) f is Borel, i.e. f-preimages of Borel (equivalently open) sets are Borel.
(2) $\mathrm{graph}(f) \in \mathcal{B}(X \times Y)$, i.e. the graph is Borel.
(3) $\mathrm{graph}(f) \in \mathbf{\Sigma}^1_1(X \times Y)$, i.e. the graph is analytic.

Proof. (1) ⇒ (2) Fix a countable basis $\{V_n\}_{n\in\mathbb{N}}$ for Y and note that $\forall (x,y)\in X\times Y$,

$$(x,y)\in \operatorname{graph}(f) \iff f(x)=y \iff \forall n\in\mathbb{N}\ (y\in V_n \implies x\in f^{-1}(V_n))$$
$$\iff \underbrace{\forall n\in\mathbb{N}}_{\text{countable }\cap}\ \big(\ \underbrace{y\notin V_n}_{\text{closed }(V_n^c)}\quad \text{or}\quad \underbrace{x\in f^{-1}(V_n)}_{\text{Borel}}\big).$$

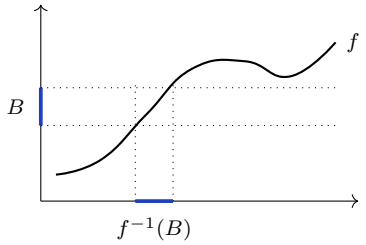
(2) ⇒ (3) is trivial.

(3) ⇒ (1) Assume $\operatorname{graph}(f)$ is analytic and fix a Borel $B\subseteq$

Source manuscript page 37

Y to show that $f^{-1}(B)$ is Borel. For each $x\in X$, we have

$$x\in f^{-1}(B) \iff \exists y\in Y\ (f(x)=y \text{ and } y\in B)$$
$$\iff \exists y\in Y\ \big[(x,y)\in\operatorname{graph}(f)\ \wedge\ (x,y)\in X\times B\big].$$



Thus

$$f^{-1}(B)=\underbrace{\operatorname{proj}_X}_{\text{continuous}}\Big[\ \underbrace{\underbrace{\operatorname{graph}(f)}_{\mathbf{\Sigma}^1_1}\cap\underbrace{(X\times B)}_{=\operatorname{proj}_Y^{-1}(B),\ \text{Borel, since } \mathcal{B} \text{ is closed under continuous preimages}}}_{\mathbf{\Sigma}^1_1 \text{ by closure under (countable) finite intersections}}\ \Big]$$

is analytic, since $\mathbf{\Sigma}^1_1$ is closed under continuous images (Proposition 4.3). But then $f^{-1}(B)^c=f^{-1}(B^c)$ is analytic, so $f^{-1}(B)\in\mathbf{\Delta}^1_1(X)=\mathcal{B}(X)$ is Borel by Souslin's theorem (Corollary 4.7). □

Corollary 4.10

Let X be Polish, $\{A_n\}_{n\in\mathbb{N}}$ a family of pairwise disjoint analytic subsets of X. Then there are pairwise disjoint Borel sets $\{B_n\}_{n\in\mathbb{N}}$ such that $B_n\supseteq A_n$, $\forall n\in\mathbb{N}$.

Proof. Exercise. Hint: Use finite intersections to construct Borel separations for each A_n, then disjointify them. □

4.4 Souslin's Operation $\mathcal{A}$

We now introduce a powerful machinery that equivalently defines the analytic sets. It further allows us to prove that all analytic sets are universally measurable, i.e. measurable with respect to an arbitrary complete σ-finite

Source manuscript page 38

measure μ that measures all Borel sets, i.e. $\mathcal{M} \supseteq \mathcal{B}(X)$, where $\mathcal{M}$ denotes the μ-measurable sets.

Definition 4.11

For a set X and a pruned tree $T \subseteq \mathbb{N}^{<\mathbb{N}}$, we call any sequence $(P_s)_{s\in T}$ of subsets of X a Souslin scheme on X. Call it monotone if $P_t \subseteq P_s$ for all $t \supseteq s$. Call it proper if it is monotone and $P_{si} \cap P_{sj} = \varnothing$, $\forall\, si, sj \in T$, $i \neq j \in \mathbb{N}$. □

Definition 4.12

Souslin's operation $\mathcal{A}$ applied to a Souslin scheme $(P_s)_{s\in T}$ on X results in the set

$$\mathcal{A}(P_s)_{s\in T} := \bigcup_{y\in[T]} \bigcap_{n\in\mathbb{N}} P_{y|_n}. \quad \square$$

Observation. Given a Souslin scheme $(P_s)_{s\in T}$, we can extend T to $\mathbb{N}^{<\mathbb{N}}$ by just setting $P_s := \varnothing$ for $s \in \mathbb{N}^{<\mathbb{N}} \setminus T$. Note that we are not assuming anything about the scheme nor about $\mathcal{A}(P_s)_{s\in T}$. Hence, we might take WLOG our $T = \mathbb{N}^{<\mathbb{N}}$. □

Proposition 4.13

If $(P_s)_{s\in T}$ is a proper Souslin scheme, then

$$\mathcal{A}(P_s)_{s\in T} = \bigcap_{n\in\mathbb{N}} \bigcup_{\substack{s\in T\\ |s|=n}} P_s.$$

Proof. $\subseteq$: If $x \in \mathcal{A}(P_s)_{s\in T}$, there is $y \in [T]$ such that $\forall n$, $x \in P_{y|_n}$. Then $\forall n$, $x \in \bigcup_{s\in T,|s|=n} P_s$.

$\supseteq$: If $x \in \bigcap_{n\in\mathbb{N}} \bigcup_{s\in T,|s|=n} P_s$, then $\forall n$, $\exists!\, s_n \in T$ with $|s_n| = n$ such that $x \in P_{s_n}$, by properness. By monotonicity, $t := s_{n+1}|_n \subseteq s_{n+1}$,

Source manuscript page 39

hence $P_t \supseteq P_{s_{n+1}} \ni x$, thus by uniqueness, $t = s_n$, and $s_n \subseteq s_{n+1}$. Thus defining $y := \bigcup_{n\in\mathbb{N}} s_n \in [T]$ we get $x \in \bigcap_{n\in\mathbb{N}} P_{y|_n} \subseteq \mathcal{A}(P_s)_{s\in T}$. □

Lemma 4.14

If a Souslin scheme $(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ satisfies $P_s = \bigcup_{i\in\mathbb{N}} P_{si}$, then $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = P_\varnothing$.

Proof. By definition, $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} \subseteq P_\varnothing$. Let $x \in P_\varnothing$; note that there is $y_0 \in \mathbb{N}$ such that $x \in P_{y_0}$, then there is $y_1 \in \mathbb{N}$ such that $x \in P_{y_0y_1}$, ... By induction we get $y \in \mathbb{N}^{\mathbb{N}}$ such that $x \in P_{y|_n}$, $\forall n \in \mathbb{N}$, so $x \in \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$. □

Definition 4.15

Let Γ be a class of sets in metrizable topological spaces. For each such space X, denote by $\mathcal{A}\Gamma(X)$ the collection of sets of the form $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ where $P_s \in \Gamma(X)$. The following shows that the operation $\mathcal{A}$ can be viewed as a projection. □

Observation. For a Souslin scheme $(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ on a set X, $\mathcal{A}(P_s)_s = \mathrm{proj}_X(P)$, where the set $P \subseteq X\times\mathbb{N}^{\mathbb{N}}$ is defined by: for each $(x,y) \in X \times \mathbb{N}^{\mathbb{N}}$,

$$\begin{aligned}(x,y)\in P \;:\iff\; & \forall n\in\mathbb{N},\; x\in P_{y|_n}\\ \iff\; & \forall n\in\mathbb{N}\;\exists s\in\mathbb{N}^{<\mathbb{N}}\;(y|_n = s \;\wedge\; x\in P_s)\\ \iff\; & \forall n\in\mathbb{N}\;\forall s\in\mathbb{N}^{<\mathbb{N}}\;(y|_n = s \implies x\in P_s).\end{aligned}$$

Moreover, if Γ is a class of sets containing all clopen sets and closed under finite unions and countable intersections (e.g. $\Gamma = \mathbf{\Pi}^0_1$ or $\mathbf{\Sigma}^1_1$), then $\mathcal{A}\Gamma \subseteq \exists^{\mathbb{N}^{\mathbb{N}}}\Gamma$ because

$$(x,y)\in P \iff \underbrace{\forall n\in\mathbb{N}}_{\text{countable}\cap}\ \underbrace{\forall s\in\mathbb{N}^{<\mathbb{N}}}_{\text{countable}\cap}\ \underbrace{\big(\underbrace{y|_n\neq s}_{\text{clopen}}\ \vee\ \underbrace{x\in P_s}_{\Gamma}\big)}_{\text{finite}\cup\,\in\,\Gamma}$$

Source manuscript page 40

and $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = \mathrm{proj}_X(P)$. In particular, $\mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \exists^{\mathbb{N}^{\mathbb{N}}}\mathbf{\Sigma}^1_1 = \mathbf{\Sigma}^1_1$, with the last equality because $\mathbf{\Sigma}^1_1$ is closed under continuous images (proj) and continuous preimages (proj^{-1}). □

We now show $\mathbf{\Sigma}^1_1 \subseteq \mathcal{A}\mathbf{\Sigma}^1_1$ in Polish spaces.

Lemma 4.16

Let X be Polish and $f : \mathbb{N}^{\mathbb{N}} \to X$ be continuous. Then for each $s \in \mathbb{N}^{<\mathbb{N}}$, $P_s := f([s])$ is analytic and

$$f(\mathbb{N}^{\mathbb{N}}) = \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = \mathcal{A}(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}}.$$

In fact, $(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ satisfies, for all $s \in \mathbb{N}^{<\mathbb{N}}$:

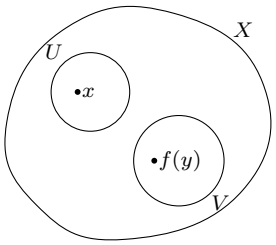
(i) $P_s \neq \varnothing$;
(ii) $P_s = \bigcup_{i\in\mathbb{N}} P_{si}$ (so that Lemma 4.14 applies);
(iii) for each $y \in \mathbb{N}^{\mathbb{N}}$ and open $U \subseteq X$, if $P_y := \bigcap_{n\in\mathbb{N}} P_{y|_n} \subseteq U$, then there is $n \in \mathbb{N}$ such that $P_{y|_n} \subseteq U$;
(iv) the scheme $(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}}$ has vanishing diameter with respect to any compatible metric on X.

Proof. We leave the verification of the analyticity of P_s and of (i) and (ii) as exercises. (ii) + Lemma 4.14 $\implies$ $\mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}} = P_\varnothing = f(\mathbb{N}^{\mathbb{N}})$. (iii) is by continuity of f: first notice that $f(y) \in f([y|_n]) = P_{y|_n}$, thus $f(y) \in P_y \subseteq U$. Hence $y \in f^{-1}(U)$, which is open, thus there is $n \in \mathbb{N}$ such that $[y|_n] \subseteq f^{-1}(U)$, i.e. $P_{y|_n} := f([y|_n]) \subseteq U$. (iv) is also by continuity since for $\varepsilon > 0$, there is $n \in \mathbb{N}$ such that $f([y|_n]) \subseteq B_\varepsilon(f(y))$, hence

$$\operatorname{diam}\overline{P_{y|_n}} = \operatorname{diam} P_{y|_n} = \operatorname{diam} f([y|_n]) \leq \operatorname{diam}\left(B_\varepsilon(f(y))\right) \leq 2\varepsilon \to 0.$$

Source manuscript page 41

It is also clear from the definition that $\mathcal{A}(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}} \supseteq \mathcal{A}(P_s)_s$, hence it remains to show the reverse inclusion. Let $x \in \mathcal{A}(\overline{P_s})_{s\in\mathbb{N}^{<\mathbb{N}}}$, so $x \in \bigcap_{n\in\mathbb{N}} \overline{P_{y|_n}}$ for some $y \in \mathbb{N}^{\mathbb{N}}$. It suffices to show $x = f(y)$, and then $f(y) \in f(\mathbb{N}^{\mathbb{N}}) = P_\varnothing = \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$. Suppose otherwise; then by Hausdorffness of X, we have disjoint open $U \ni x$, $V \ni f(y)$, thus $x \notin \overline{V}$. By continuity of f, there is $n \in \mathbb{N}$ such that $f([y|_n]) \subseteq V$, hence $x \notin \overline{f([y|_n])} = \overline{P_{y|_n}} \subseteq \overline{V}$, a contradiction.



□

Theorem 4.17 (Characterization of Σ^1_1 via $\mathcal{A}$)

Let X be Polish and $\varnothing \neq A \subseteq X$. The following are equivalent:

(1) A is analytic ($\iff$ there is a continuous $f : \mathbb{N}^{\mathbb{N}} \to X$ such that $f(\mathbb{N}^{\mathbb{N}}) = A$).
(2) $A = \mathcal{A}(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$, each $F_s \neq \varnothing$ closed and the scheme $(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$ monotone with vanishing diameter.
(3) $A = \mathcal{A}(P_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$, each P_s analytic.

As a result,

$$\mathbf{\Sigma}^1_1 = \mathcal{A}\mathbf{\Sigma}^1_1 = \mathcal{A}\mathbf{\Pi}^0_1.$$

Proof. (1) $\Rightarrow$ (2): Take $f : \mathbb{N}^{\mathbb{N}} \to X$ continuous with $A = f(\mathbb{N}^{\mathbb{N}})$. Let $F_s := \overline{f([s])}$. Then Lemma 4.16 gives $A = \mathcal{A}(F_s)_{s\in\mathbb{N}^{<\mathbb{N}}}$.

(2) $\Rightarrow$ (3): Trivial since closed sets are analytic.

(3) $\Rightarrow$ (1): By the observation above, $\mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \mathbf{\Sigma}^1_1$.

For the last claim, note that $\mathcal{A}\mathbf{\Pi}^0_1 \subseteq \mathcal{A}\mathbf{\Sigma}^1_1$ since closed sets are analytic, while (1) $\Rightarrow$ (2) gives $\mathbf{\Sigma}^1_1 \subseteq \mathcal{A}\mathbf{\Pi}^0_1$ (with $\varnothing = \mathcal{A}(\varnothing)_{s\in\mathbb{N}^{<\mathbb{N}}}$) and the observation gives $\mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \mathbf{\Sigma}^1_1$. Hence

$$\mathbf{\Sigma}^1_1 \subseteq \mathcal{A}\mathbf{\Pi}^0_1 \subseteq \mathcal{A}\mathbf{\Sigma}^1_1 \subseteq \mathbf{\Sigma}^1_1,$$

so all three classes coincide; in particular $\mathcal{A}\mathbf{\Sigma}^1_1 = \mathcal{A}\mathbf{\Pi}^0_1$. □